



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering.

Academic Year: 2023 - 2024(Odd/Even Semester)

Degree, Semester & Branch: B.E, V & CSE

Course Code & Title: CS3551& Distributed Computing

Name of the Faculty member (s): A. Sofia Thanga Mary, AP/CSE

Innovative Practice Description

- **Unit / Topic: Unit IV/** Check pointing and rollback recovery
- **Course Outcome: CO 4**
- **Topic Learning Outcome: TLO9**
- **Activity Chosen:** Flipped class Room
- **Justification:**

A flipped classroom is a type of collaborative learning where students might watch videos took the notes according to their understanding level at home and then do the task provided in classroom with gained knowledge. This improves the habit of self-learning among the students.
- **Time Allotted for the Activity: 40 Minutes**
- **Details of the Implementation:**
 - Flipping the classroom is an inverting the classroom approach to teaching. In this approach, the traditional in-class teaching is “flipped” to better meet the needs of individual learners. Students gain control of the learning process through studying course material outside of class, using readings, pre-recorded video lectures. It helps the faculty/lecturer to redefine in-class activities and include homework problems and keep the students engaged in the content.
 - **Identify and Share:** Related materials are identified and I posted relevant references/materials/notes of Check pointing and rollback recovery to the students and given as self-learning.
 - **Evaluate:** I have prepared few questions related to the content shared and ask those questions in next class session and make the students to write/present in class. The Figure.1 and Figure 2 images shows the student Shamista and MuthuSamy explaining the self-explored concepts on check pointing and recovery. Make the students to find out the answer by their own by learning.

- Images / Screenshot of the practice:



Fig 1: Shamista's presentation about check pointing and recovery



Fig 2: Muthusamy's presentation about Check pointing and recovery

- CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO10	PO12	PSO1	PSO2	PSO3
C304.4	3	2	1	1	2	2	3	3	2

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO10
	3	2	1	1	1
Justification for correlation	Students will be able to use the basic mathematical knowledge, to solve consensus and agreement problem	Students will be able to identify, formulate and analyze the best solution for problem with the knowledge of the check pointing, rollback and consensus and agreement algorithm	Students could be able to provide the solution for complex problem with the knowledge of consensus and agreement concepts.	Students will be able to solve the simple research problem by using the knowledge of recovery and consensus.	The students could be able to explain the consensus and agreement problem.

PO12	PSO1	PSO2	PSO3
2	3	3	2
Students could be able to continuously improve their knowledge in cloud computing concepts by completing some online course.	Students will be able to use the check pointing and consensus mechanism to develop the distributed mobile software.	Students could be able to provide the distribute solution with the knowledge of check pointing mechanism	Students could be able to use the concept of check pointing, rollback and consensus mechanism in AI and big data analytics.

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- Student felt good, since they can study at their own pace/time.
- They felt that through such learning, they can explore more.

❖ *Benefit of the practice:*

- More one-to-one time between teacher and student.
- More collaboration time for students.
- Students learn at their own pace.
- Practical things – like missing class due to illness – become less problematic.
- It encourages students to come to class prepared.

❖ *Challenges faced in implementation:*

- Relies on student preparation – few students did not come prepared.
- The depth of the subject can be dictated by the student themselves or the group the student is working with.
- The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a traditional class.

References:

- ❖ <https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model>
- ❖ https://en.wikipedia.org/wiki/Flipped_classroom
- ❖ [https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162- what-why-and-how-to-implement-a-flipped-classroom-model](https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model)
- ❖ <https://educationaltechnologyjournal.springeropen.com/articles/10.1186/s41239-016-0032-z>

Signature of Faculty Member

HOD